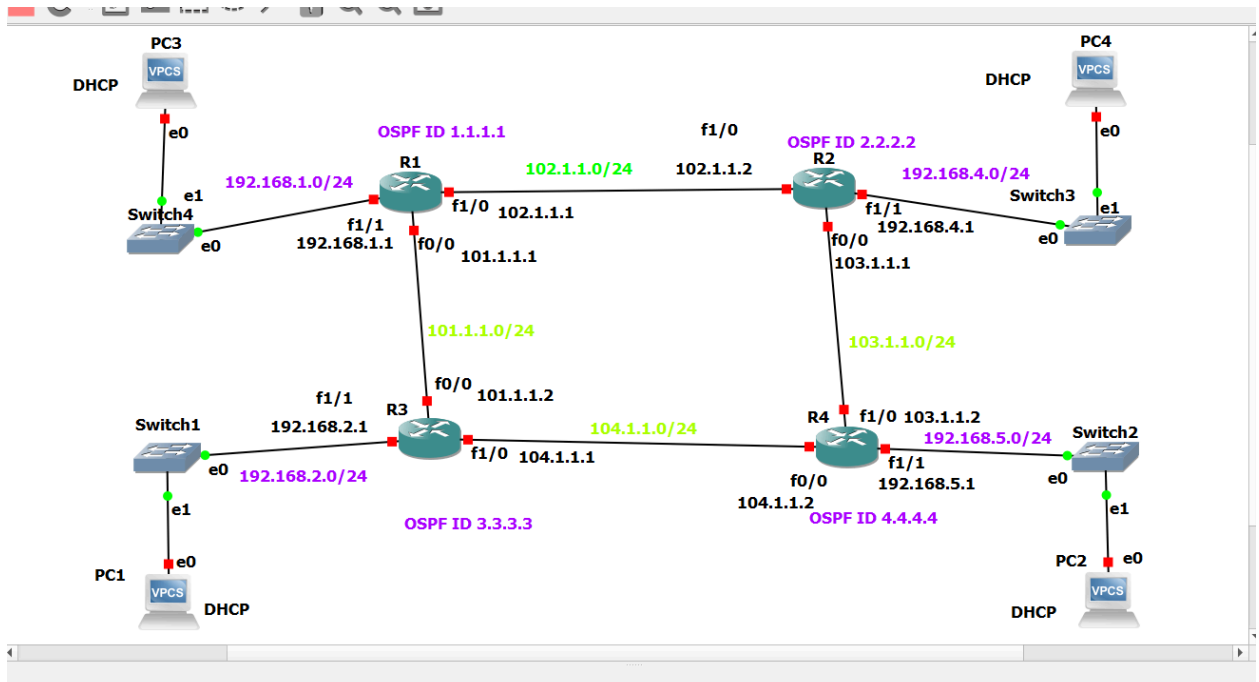


1. Brian Wei



2.

```
PC1> ping 192.168.5.2

84 bytes from 192.168.5.2 icmp_seq=1 ttl=62 time=54.273 ms
84 bytes from 192.168.5.2 icmp_seq=2 ttl=62 time=42.498 ms
84 bytes from 192.168.5.2 icmp_seq=3 ttl=62 time=41.780 ms
84 bytes from 192.168.5.2 icmp_seq=4 ttl=62 time=42.085 ms
84 bytes from 192.168.5.2 icmp_seq=5 ttl=62 time=36.519 ms

PC1>
PC1> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=62 time=48.000 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=62 time=36.339 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=62 time=34.858 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=62 time=40.120 ms
^C
PC1> ping 192.168.4.2

84 bytes from 192.168.4.2 icmp_seq=1 ttl=61 time=66.704 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=61 time=61.358 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=61 time=39.628 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=61 time=48.432 ms
^C
PC1> █
```

3.

```
PC2> ping 192.168.2.2
```

```
84 bytes from 192.168.2.2 icmp_seq=1 ttl=62 time=36.736 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=62 time=43.557 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=62 time=22.896 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=62 time=41.085 ms
^C
```

```
PC2> ping 192.168.1.2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=61 time=51.041 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=61 time=53.938 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=61 time=62.902 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=61 time=47.357 ms
^C
```

```
PC2> ping 192.168.4.2
```

```
84 bytes from 192.168.4.2 icmp_seq=1 ttl=62 time=43.560 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=62 time=42.923 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=62 time=38.143 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=62 time=43.656 ms
^C
```

4.

```
PC3> ping 192.168.2.2
```

```
84 bytes from 192.168.2.2 icmp_seq=1 ttl=62 time=51.298 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=62 time=36.866 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=62 time=36.333 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=62 time=26.306 ms
^C
```

```
PC3> ping 192.168.5.2
```

```
84 bytes from 192.168.5.2 icmp_seq=1 ttl=61 time=54.444 ms
84 bytes from 192.168.5.2 icmp_seq=2 ttl=61 time=55.334 ms
84 bytes from 192.168.5.2 icmp_seq=3 ttl=61 time=37.891 ms
84 bytes from 192.168.5.2 icmp_seq=4 ttl=61 time=34.626 ms
^C
```

```
PC3> ping 192.168.4.2
```

```
84 bytes from 192.168.4.2 icmp_seq=1 ttl=62 time=45.744 ms
84 bytes from 192.168.4.2 icmp_seq=2 ttl=62 time=33.833 ms
84 bytes from 192.168.4.2 icmp_seq=3 ttl=62 time=35.062 ms
84 bytes from 192.168.4.2 icmp_seq=4 ttl=62 time=32.615 ms
^C
```

5.

```
PC3> █
```

```
PC4> ping 192.168.2.2
```

```
84 bytes from 192.168.2.2 icmp_seq=1 ttl=61 time=53.722 ms
84 bytes from 192.168.2.2 icmp_seq=2 ttl=61 time=60.011 ms
84 bytes from 192.168.2.2 icmp_seq=3 ttl=61 time=52.867 ms
84 bytes from 192.168.2.2 icmp_seq=4 ttl=61 time=60.551 ms
^C
```

```
PC4> ping 192.168.5.2
```

```
84 bytes from 192.168.5.2 icmp_seq=1 ttl=62 time=38.250 ms
84 bytes from 192.168.5.2 icmp_seq=2 ttl=62 time=33.464 ms
84 bytes from 192.168.5.2 icmp_seq=3 ttl=62 time=24.810 ms
84 bytes from 192.168.5.2 icmp_seq=4 ttl=62 time=42.377 ms
^C
```

```
PC4> ping 192.168.1.2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=62 time=40.585 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=62 time=32.561 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=62 time=27.210 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=62 time=24.948 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=62 time=36.519 ms
^C
```

```
PC4> █
```

6.

```
R1#show ip ospf database
```

```
OSPF Router with ID (1.1.1.1) (Process ID 1)
```

```
Router Link States (Area 0)
```

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	31	0x80000003	0x004780	5
2.2.2.2	2.2.2.2	34	0x80000003	0x002493	5
3.3.3.3	3.3.3.3	32	0x80000005	0x00911E	5
4.4.4.4	4.4.4.4	51	0x80000003	0x00ABF5	5

```
R1#
```

```
R2#show ip ospf database
```

```
OSPF Router with ID (2.2.2.2) (Process ID 1)
```

```
Router Link States (Area 0)
```

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	110	0x80000003	0x004780	5
2.2.2.2	2.2.2.2	112	0x80000003	0x002493	5
3.3.3.3	3.3.3.3	111	0x80000005	0x00911E	5
4.4.4.4	4.4.4.4	130	0x80000003	0x00ABF5	5

```
R2#
```

FROM LOADING TO FULL, Loading Done

R3#show ip ospf database

OSPF Router with ID (3.3.3.3) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	152	0x80000003	0x004780	5
2.2.2.2	2.2.2.2	155	0x80000003	0x002493	5
3.3.3.3	3.3.3.3	151	0x80000005	0x00911E	5
4.4.4.4	4.4.4.4	172	0x80000003	0x00ABF5	5

R3#

FROM LOADING TO FULL, Loading Done

R4#show ip ospf database

OSPF Router with ID (4.4.4.4) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	181	0x80000003	0x004780	5
2.2.2.2	2.2.2.2	183	0x80000003	0x002493	5
3.3.3.3	3.3.3.3	180	0x80000005	0x00911E	5
4.4.4.4	4.4.4.4	199	0x80000003	0x00ABF5	5

R4#

The output shows the OSPF router id that was set up during configuration. It also shows the process ID which is 1 for all of them. It also lists the router link states for the area which is area 0 for all of them. Next, it lists the link id. And the ADV router is listed which is the router that started the Link State Advertisement. It also lists the age to show how old the link is. The sequence number is associated with the link state advertisement. Checksum is listed so that it can be compared to see if everything is good. The link count states the number of links that the router has.